

GRADE 6 MATHEMATICS

Classroom Observation Guide

Grade 6 Mathematics — Day Four: “Testing the Model”

Professor Xavier Honablue, M.Ed.

Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Today marks the first full, multi-board implementation of the Engage-Explore-Enrich (E3) station rotation model in this classroom. Rather than a single front board, multiple smart boards run simultaneously, and students rotate individually rather than as a whole class. This guide explains the design of today's pilot, the technology safeguards in place, the standards addressed, and the research behind the instructional choices.

Because this is a first implementation, observers should expect some logistical adjustment as the rotation is refined in real time.

Lesson at a Glance (55 Minutes)

Time	Segment	What's Happening
0-3 min	Welcome	Students are reminded that yesterday they were on stage performing skits, and told that today they are “on stage” again in a quieter way - through one-on-one Enrich conversations - as the class helps test the first full multi-board E3 rotation.
3-8 min	Engage	Front-board introduction to IXL.com and today's exploratory purpose - browsing, not mastering.
8-11 min	The Rules	Paper and pencil required at all times; no calculators; all websites except IXL are blocked by Blocksi.
11-14 min	Rotation Explained	Overview of the three concurrent positions: Explore, While-You-Wait practice, and Enrich.
14-45 min	Rotation (Explore / Wait / Enrich)	Students rotate individually across smart boards: browsing and journaling IXL topics, redoing prior work slowly while waiting, and receiving one-on-one tutoring at the

45-49 min	Your Choice Becomes Your Syllabus	Enrich board. Students learn their chosen IXL topic will be incorporated into their individual grade 6 syllabus.
49-52 min	Turn It In	Journals are collected for teacher review before the next class.
52-55 min	What You Just Did	Standards are named in student-friendly language; students journal which topic they would choose first, and why.

The Rotation Model, Explained

Today's rotation places each student in one of three positions at any given moment, made possible by running several smart boards simultaneously rather than a single shared board:

- **Explore:** Students browse grade 6 topics on IXL.com at their assigned smart board and record, in a paper journal, which topics interest them and why.
- **While You Wait:** Rather than idling, students pull out previously completed work and redo it slowly and carefully - deliberate, precision-focused practice rather than downtime.
- **Enrich:** Students bring their journal to Professor Honablu for one-on-one or small-group tutoring on the topic they selected.

Technology Safeguards

Paper and pencil are required at every station; calculators are not permitted, so that conceptual reasoning - not computation shortcuts - drives today's exploration. All classroom devices are filtered through Blocks, a web-filtering platform, so that IXL.com is the only accessible site during this activity.

From Exploration to Personalized Syllabus

The topics each student journals today are intended to directly inform that student's individual grade 6 syllabus going forward. In practice, this means two students in the same classroom may follow different topic sequences later in the year, based on interests identified today.

Common Core & Michigan Standards Addressed

Mathematical Practices

- **MP1:** Make sense of problems and persevere in solving them - choosing a topic, then working through it at the Enrich board.
- **MP5:** Use appropriate tools strategically - paper and pencil in place of a calculator, by design.
- **MP6:** Attend to precision - the While-You-Wait slow redo is deliberate precision practice, not filler.

Content Standards

Because each student selects their own IXL topic, the specific grade 6 content strand addressed - Ratios & Proportional Relationships, the Number System, Expressions & Equations, Geometry, or Statistics & Probability - varies by student. Today functions as a diagnostic and interest-mapping session that shapes each student's forthcoming individualized syllabus, rather than delivering a single shared content standard to the whole class.

Pedagogical & Cognitive Research Foundations

Student Choice and Self-Determination Theory

Allowing students to select their own topic of interest is grounded in Self-Determination Theory (Ryan & Deci, 2000), which identifies autonomy - along with competence and relatedness - as a core driver of intrinsic motivation. Autonomy-supportive teaching, which offers genuine choice, is associated with deeper engagement and more effective learning than externally controlled assignments.

Personalized Learning

Building each student's forthcoming syllabus around topics they identify as interesting reflects the broader personalized-learning movement in K-12 education, in which instructional content and pacing are adapted to individual student needs and interests rather than delivered uniformly (Pane, Steiner, Baird, & Hamilton, 2015).

Deliberate Practice

The “While You Wait” station is grounded in deliberate practice research (Ericsson, Krampe, & Tesch-Römer, 1993), which found that structured, effortful, and precision-focused practice - not mere repetition - is a key driver of skill development. Redoing a problem slowly and carefully is treated as practice in its own right, not as a placeholder activity. (Later replications have found smaller effects than the original study; the framework is used here as a rationale for precision-focused practice, not as a claim that practice alone determines all differences in student outcomes.)

Continued Foundations

Today continues the gradual release of responsibility (Pearson & Gallagher, 1983) established on Day One: the front-board Engage segment models the tool and the rules, before responsibility shifts to independent Explore work and then to individualized Enrich support.

What to Look For as an Observer

- Whether students are genuinely exploring a range of IXL topics, or defaulting to the first topic they see - variety in journal entries is a sign of authentic exploration.
- The quality of the “While You Wait” redo work - look for visibly slower, more deliberate handwriting and work compared to a student's original attempt, not just repetition.
- How Enrich conversations differ from student to student - since each is centered on an individually chosen topic, no two Enrich conversations should look identical.
- General smoothness of the rotation itself - as a first implementation, note any bottlenecks (e.g., wait times, board access) worth adjusting before the next run.

Appendix: Day Four in the Grade 6 Math Lesson Plan Template

For administrators cross-referencing today's observation against the school's required weekly lesson plan format, the table below shows how today's lesson populates that same template.

Template Field	Day Four Content
Standard(s) & Taxonomy Level	MP1, MP5, MP6; content standard varies by student (personalized) C/A
Learning Target(s) (SWBAT)	SWBAT explore a range of grade 6 math topics and identify a personal area of interest; engage in precision-focused practice while waiting; and articulate a chosen topic during one-on-one instruction.
Success Criteria (I Can)	I can explore a set of math topics and identify which ones genuinely interest me. I can slow down and redo my own work carefully, like a professional would. I can advocate for what I want to learn.
Daily Marzano Strategies	Helping Students Engage in Cognitively Complex Tasks; Helping Students Practice Skills, Strategies, and Processes; Identifying Critical Content
Do-Now (5 min)	Students are told today is the first full test of the multi-board E3 rotation model.
Instructional Plan	Front-board Engage introduces IXL.com and the day's rules. Students rotate individually across three positions: Explore (browse and journal IXL topics), While You Wait (redo prior work slowly and precisely), and Enrich (one-on-one tutoring on a chosen topic with Professor Honablue). Chosen topics are collected to inform each student's personalized syllabus.
Critical Content Debrief (5 min)	Can students identify a specific grade 6 topic of genuine interest and articulate why it interests them, rather than defaulting to an arbitrary choice?
Daily Formative [Ungraded] / Summative [Graded] Assessments	Formative: teacher observation during Explore and While-You-Wait; one-on-one Enrich conversation. Summative: none today - collected journals serve as the basis for building each student's individualized syllabus.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
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Clear modeling of IXL navigation and journaling expectations before rotation begins; visible, consistent rules posted at every station (paper/pencil, no calculators, IXL only).

A curated shortlist of IXL topics for students who feel overwhelmed by the full topic menu; sentence starters for journaling ("This topic interests me because...").

Priority Enrich rotation slots for students needing more one-on-one time; a paraprofessional or peer partner for students who need support navigating IXL independently.

Data Source: Classroom Formative Assessments [Ungraded]; student journals.

Explore More: Resources for Observers

For guests who would like to look deeper into the frameworks behind today's model and tools.

Self-Determination Theory

Ryan and Deci's overview of autonomy, competence, and relatedness as drivers of intrinsic motivation - the basis for today's student-choice design.

[View on selfdeterminationtheory.org](https://selfdeterminationtheory.org)

Standards for Mathematical Practice

The official Common Core description of MP1, MP5, MP6, and the other Standards for Mathematical Practice referenced throughout this unit.

[View on corestandards.org](https://corestandards.org)

IXL.com

The adaptive practice platform students explored today, covering the full grade 6 mathematics topic range.

[View Grade 6 Math on IXL](#)

Michigan K-12 Math Standards

The Michigan Department of Education's full K-12 Standards for Mathematics document.

[View Michigan DOE PDF](#)

References

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363-406.

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Pearson, P. D., & Gallagher, M. C. (1983). The instruction of reading comprehension. *Contemporary Educational Psychology*, 8(3), 317-344.

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